

Chapter 6 Homework

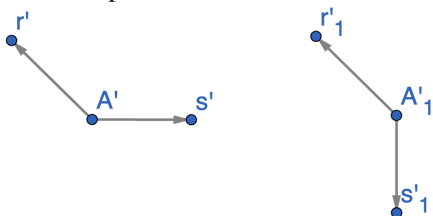
Donald Timpson

December 10, 2024

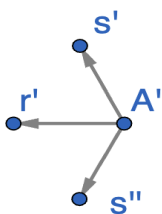
Exercise Group 6.1

1) If $\angle A$ is not straight then there's two, otherwise there's one.

2) The two options are drawn below.



3) The potential s' and s'' are drawn below.



4) There is only one such angle.



5) Yes. Any angle congruent to a straight angle is a straight angle.

Exercise Group 6.2

1) a) Every angle is congruent with itself.

b) If angle A is congruent to angle B , then angle B is congruent to angle A .

c) If angle A is congruent to angle B and angle B is congruent to angle C , then angle A is congruent to angle C .

2) *Proof.* If $\angle E \cong \angle F$, $\angle F \cong \angle G$, and $\angle G \cong \angle H$, then $\angle E \cong \angle H$.

1. Assume $\angle E \cong \angle F$, $\angle F \cong \angle G$, and $\angle G \cong \angle H$.

2. Since $\angle E \cong \angle F$ and $\angle F \cong \angle G$, then by the transitive property $\angle E \cong \angle G$.

3. Since $\angle E \cong \angle G$ and $\angle G \cong \angle H$, then by the transitive property $\angle E \cong \angle H$. □

3) *Proof.* If $\angle R \cong \angle S$ and $\angle T \cong \angle S$, then $\angle R \cong \angle T$.

1. Assume $\angle R \cong \angle S$ and $\angle T \cong \angle S$.

2. Since $\angle T \cong \angle S$, then by the symmetric property $\angle S \cong \angle T$.

3. Since $\angle R \cong \angle S$ and $\angle S \cong \angle T$, then by the transitive property $\angle R \cong \angle T$. □

- 4) Use Postulate V-1 and (b) and (c) of Postulate V-2 to prove that for every $\angle A$, $\angle A \cong \angle A$.

Proof. For every $\angle A$, $\angle A \cong \angle A$.

1. Let $\angle A$ be any angle, let r' be a ray from A' on a line l' .
2. Then by V-1, there exists a ray s' from the point A' s.t. $\angle A' \cong \angle A$.
3. Since $\angle A' \cong \angle A$, then by V-2 (b) $\angle A \cong \angle A'$.
4. Since $\angle A \cong \angle A'$ and $\angle A' \cong \angle A$, then by V-2 (c) $\angle A \cong \angle A$.

□



Exercise Group 6.3

- 1)
 - a) 85°
 - b) 116°
 - c) 150°
 - d) Yes
 - e) 54°
- 2) $\angle POQ \cong \angle SVT$
- 3) $\angle BAD \cong \angle EAC$
- 4) $\angle RPS \cong \angle UPT$
- 5) 2: Subtracting, 3: Adding, 4: Subtracting

Exercise Group 6.4

- 1) A right angle.
- 2) *Proof.* If $\angle 1 \cong \angle 2$, then $\angle EAC \cong \angle CDA$.
 1. Assume $\angle 1 \cong \angle 2$, and observe that $\angle DAC \cong \angle CAD$.
 2. Then by V-3, $\angle EAC \cong \angle CDA$.
- 3) *Proof.* $\angle A \cong \angle E$.
 1. Observe that $\angle A$ is congruent to its vertical angle $\angle B$ by Thm 6-2.
 2. Let $\angle F$ be the supplement of $\angle C$, and observe that $\angle B \cong \angle F$ is given.
 3. Since $\angle C \cong \angle D$, then by Thm 6-1 $\angle F \cong \angle E$.
 4. Since $\angle B \cong \angle F$ and $\angle F \cong \angle E$, then $\angle B \cong \angle E$ by the transitive property.
 5. Finally, since $\angle A \cong \angle B$ and $\angle B \cong \angle E$, then $\angle A \cong \angle E$ by the transitive property.
- 4) An angle is congruent to a right angle. Show that it, too, is a right angle.

□

□

Proof. An angle congruent to a right angle is a right angle.

1. Let $\angle A$ be a right angle, let $\angle B \cong \angle A$, let $\angle A'$ be the supplement of $\angle A$, and let $\angle B'$ be the supplement of $\angle B$.
2. Since $\angle A$ is a right angle, then $\angle A \cong \angle A'$.
3. Since $\angle A \cong \angle B$, then by Thm 6-1 their supplements are congruent, and so $\angle A' \cong \angle B'$.
4. Then $\angle B \cong \angle A \cong \angle A' \cong \angle B'$, and so $\angle B \cong \angle B'$.
5. Therefore $\angle B$ is a right angle (it's congruent to its supplement).

□

5) *Proof.* If CD bisects $\angle AOB$ in Fig. 6-15, then it also bisects $\angle EOF$.

1. Since CD bisects $\angle AOB$, then $\angle AOC \cong \angle BOC$.

2. Since vertical angles are congruent, then $\angle EOD \cong \angle BOC$ and $\angle FOD \cong \angle AOC$.

3. Then $\angle EOD \cong \angle BOC \cong \angle AOC \cong \angle FOD$.

4. Since $\angle EOD \cong \angle FOD$, then CD bisects $\angle EOF$. □

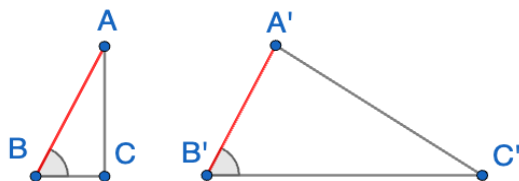
6) Perpendicular: relation, parallel: relation, vertex: relation, altitude: relation, greater than: relation, angle: property, segment: property, median: relation, right (angle): property, straight (angle): property, bisector: relation.

Exercise Group 6.5

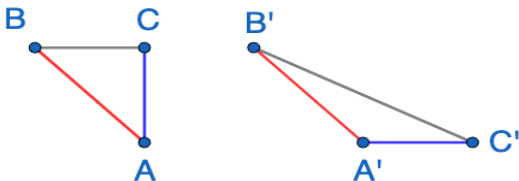
1) $\angle C \cong \angle E$ and $\angle A \cong \angle F$.

2) $PQ \cong SU$, $\angle P \cong \angle S$, $PR \cong ST$, $\angle R \cong \angle T$, and $QR \cong UT$.

3) Not that $\angle C \cong \angle C'$.



4) Not that $\angle B \cong \angle B'$.



5) *Proof.* In figure 6-19, if $AC \cong AD$ and $\angle CAB \cong \angle BAD$, then $\angle C \cong \angle D$.

1. Assume $AC \cong AD$ and $\angle CAB \cong \angle DAB$.

2. Since $AB \cong AB$, then $\angle C \cong \angle D$ by S.A.S. postulate. □

6) *Proof.* In figure 6-20, if $DO \cong OB$ and $AO \cong OC$, then $\angle A \cong \angle C$.

1. Assume $DO \cong OB$ and $AO \cong OC$.

2. Observe that $\angle AOD \cong \angle COB$ since they're vertical angles.

3. Since $AO \cong CO$, $\angle AOD \cong \angle COB$, and $OD \cong OB$, then $\angle A \cong \angle C$ by S.A.S. postulate. □

7) *Proof.* In figure 6-21, if $AB \cong CD$ and $\angle CBA \cong \angle BCD$, then $\angle CAB \cong \angle BDC$.

1. Assume $AB \cong CD$ and $\angle CBA \cong \angle BCD$.

2. Observe that $BC \cong CB$.

3. Since $AB \cong DC$, $\angle ABC \cong \angle DCB$, and $BC \cong CB$, then $\angle CAB \cong \angle BDC$ by S.A.S postulate. □

8) *Proof.* In figure 6-22, if $AD \cong BD$ and $\angle ADC$ is a right angle, then $\angle A \cong \angle B$.

1. Assume $AD \cong BD$ and $\angle ADC$ is a right angle.

2. Since $\angle ADC$ is a right angle, then $\angle ADC \cong \angle BDC$.

3. Then $AD \cong BD$, $\angle ADC \cong \angle BDC$, and $DC \cong DC$, so $\angle A \cong \angle B$ by S.A.S postulate. □

9) *Proof.* In figure 6-23, if $\angle EBC \cong \angle ABD$, $AB \cong BC$, and $BE \cong BD$, then $\angle A \cong \angle C$.

1. Assume $\angle EBC \cong \angle ABD$, $AB \cong BC$, and $BE \cong BD$.

2. Since $\angle EBC \cong \angle ABD$ and $\angle EBD \cong \angle DBE$, then $\angle DBC \cong \angle EBA$.

3. Since $DB \cong EB$, $\angle DBC \cong \angle EBA$, and $BC \cong BA$, then $\angle A \cong \angle C$ by S.A.S. postulate. □

10) *Proof.* In figure 6-24, if $AD \cong DC$, $\angle 1 \cong \angle C$, and $\angle 2 \cong \angle C$, then $\angle 3 \cong \angle 4$.

1. Assume $AD \cong DC$, $\angle 1 = \angle CDB \cong \angle C$, and $\angle 2 = \angle ADB \cong \angle C$.

2. Since $AD \cong CD$, $\angle ADB \cong \angle C \cong \angle CDB$, and $DB \cong DB$, then $\angle 3 = \angle ABD \cong \angle CBD = \angle 4$ by S.A.S. postulate. □

- 11) *Proof.* In figure 6-25, if $\angle 1 \cong \angle 2$, $\angle BCF \cong \angle ADF$, $CG \cong HD$, and $CF \cong DF$, then $\angle 3 \cong \angle 4$.
1. Assume $\angle 1 \cong \angle 2$, $\angle BCF \cong \angle ADF$, $CG \cong HD$, and $CF \cong DF$.
 2. Since $\angle BCF \cong \angle ADF$, $\angle BCG = \angle 1 \cong \angle 2 = \angle ADH$, then $\angle GCF \cong \angle HDF$.
 3. Since $GC \cong HD$, $\angle GCF \cong \angle HDF$, and $CF \cong DF$, then $\angle 3 = \angle GFC \cong \angle HFD = \angle 4$ by S.A.S. postulate. $\square$
- 12) *Proof.* In figure 6-26, if $\angle 1 \cong \angle 2$, $AE \cong CD$, and $AB \cong BC$, then $\angle E \cong \angle D$.
1. Assume $\angle 1 \cong \angle 2$, $AE \cong CD$, and $AB \cong BC$.
 2. Observe that $\angle EAB \cong \angle 1 \cong \angle 2 \cong \angle DCB$.
 3. Since $EA \cong DC$, $\angle EAB \cong \angle DCB$, and $AB \cong CB$, then $\angle E \cong \angle D$ by S.A.S. postulate. $\square$
- 13) *Proof.* In figure 6-27, if $\angle 1 \cong \angle 2$, $AC \cong BC$, and $AM \cong BM$, then $\angle 3 \cong \angle 4$.
1. Assume $\angle 1 \cong \angle 2$, $AC \cong BC$, and $AM \cong BM$.
 2. Observe that since $\angle 1 \cong \angle 2$, then their supplementary angles $\angle CAB$ and $\angle CBM$ are congruent.
 3. Since $CA \cong CB$, $\angle CAM \cong \angle CBM$, and $AM \cong BM$, then $\angle 3 = \angle ACM \cong \angle BCM = \angle 4$ by S.A.S. postulate. $\square$
- 14) *Proof.* In figure 6-28, if $\angle ADC \cong \angle ABC$, $\angle 1 \cong \angle 2$, and $AD \cong BC$, then $\angle A \cong \angle C$.
1. Assume $\angle ADC \cong \angle ABC$, $\angle 1 \cong \angle 2$, and $AD \cong BC$.
 2. Since $\angle ADC \cong \angle ABC$ and $\angle BDC = \angle 1 \cong \angle 2 = \angle DBA$, then $\angle ADB \cong \angle CBD$.
 3. Since $AD \cong CB$, $\angle ADB \cong \angle CBD$, and $DB \cong BD$, then $\angle A \cong \angle C$ by S.A.S. postulate. $\square$
- 15) *Proof.* In figure 6-29, if $AD \cong DC$ and $\angle 1 \cong \angle 2$, then $\angle A \cong \angle C$.
1. Assume $AD \cong DC$ and $\angle 1 \cong \angle 2$.
 2. Since $\angle 1 \cong \angle 2$, then their supplementary angles $\angle CDB$ and $\angle ADB$ are congruent.
 3. Since $AD \cong CD$, $\angle ADB \cong \angle CDB$, and $DB \cong DB$, then $\angle A \cong \angle C$ by S.A.S. postulate. $\square$
- 16) *Proof.* In figure 6-30, if $BF \cong ED$, $AE \cong CF$, and $\angle 1 \cong \angle 2$, then $\angle 3 \cong \angle 4$ and $\angle 5 \cong \angle 6$.
1. Assume $BF \cong ED$, $AE \cong CF$, and $\angle 1 = \angle AFB \cong \angle CED = \angle 2$.
 2. Observe that since $AE \cong CF$ and $FE \cong EF$, then $AF \cong CE$.
 3. Since $AF \cong CE$, $\angle AFB \cong \angle CED$, and $FB \cong ED$, then $\angle 3 = \angle BAF \cong \angle DCE = \angle 4$ by S.A.S. postulate.
 5. Now, since $\angle AFE \cong \angle CED$, then their supplementary angles $\angle BFC$ and $\angle DEA$ are congruent.
 6. Since $AE \cong CF$, $\angle AED \cong \angle CFB$, and $ED \cong FB$, then $\angle 5 = \angle BCF \cong \angle DAE = \angle 6$ by S.A.S. postulate. $\square$
- 17) *Proof.* In figure 6-31, if $AB \cong AD$, $BO \cong OD$, and $\angle OAB \cong \angle OAD$, then $\angle 1 \cong \angle 2$.
1. Assume $AB \cong AD$, $BO \cong OD$, and $\angle OAB \cong \angle OAD$.
 2. Since $OA \cong OA$, $\angle OAB \cong \angle OAD$, and $AB \cong AD$, then $\angle BOA \cong \angle DOA$ by S.A.S. postulate.
 3. Since $\angle BOA \cong \angle DOA$, then their supplementary angles $\angle BOC$ and $\angle DOC$ are congruent.
 4. Since $BO \cong DO$, $\angle BOC \cong \angle DOC$, and $OC \cong OC$, then by S.A.S. postulate $\angle 1 = \angle OBC \cong \angle ODC = \angle 2$. $\square$

Exercise Group 6.6

- 1) Using Thm 6-3, prove that an equilateral triangle is equiangular.

Proof. If a triangle is equilateral, then it is equiangular.

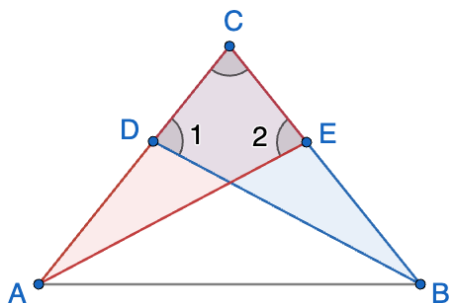
1. Assume $\triangle ABC$ is an equilateral triangle.
2. Since $AC \cong BC$, then by Thm 6-3 $\angle A \cong \angle B$.
3. Since $BA \cong CA$, then by Thm 6-3 $\angle B \cong \angle C$.
4. Then $\angle \cong \angle B \cong \angle C$.
5. Therefore $\triangle ABC$ is an equiangular triangle. $\square$

- 2) I believe this problem has a typo, and that it should say "If $PQ \cong PR$, and $\angle Q$ is 44° ..."

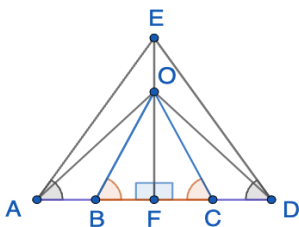
Proof. In figure 6-33, if in $\triangle PQR$ we have $PQ \cong PR$ and $\angle Q$ is 44° , then $\angle R$ is 44° .

1. Let $\triangle PQR$ where $PQ \cong PR$ and $\angle Q$ is 44° .
2. Then $\triangle PQR$ is an isosceles triangle, and so by Thm 6-3 $\angle Q \cong \angle R$.
3. Since $\angle Q$ is 44° , then $\angle R$ is 44° . $\square$

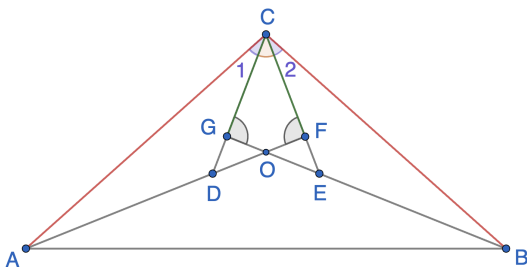
- 3) *Proof.* In figure 6-34, if $AB \cong AC$ and $BD \cong DC$, then $\angle ABC \cong \angle ACB$, $\angle CBD \cong \angle BCD$, and $\angle ABD \cong \angle ACD$.
1. Assume $AB \cong AC$ and $BD \cong DC$.
 2. Since $AB \cong AC$ then $\triangle BAC$ is isosceles, and since $BD \cong DC$ then $\triangle BDC$ is isosceles.
 3. Since $\angle ABC$ and $\angle ACB$ are base angles of $\triangle BAC$, then $\angle ABC \cong \angle ACB$.
 4. Since $\angle CBD$ and $\angle BCD$ are base angles of $\triangle BDC$, then $\angle CBD \cong \angle BCD$.
 5. Since $\angle ABC \cong \angle ACB$ and $\angle CBD \cong \angle BCD$, then $\angle ABD \cong \angle ACD$. □
- 4) *Proof.* In figure 6-35, if $AC \cong BC$ and $AD \cong BD$, then $\angle 1 \cong \angle 2$.
1. Assume $AC \cong BC$ and $AD \cong BD$.
 2. Since $AC \cong BC$, then $\triangle ACB$ is an isosceles triangle with base angles $\angle CAD \cong \angle CBD$.
 3. Since $CA \cong CB$, $\angle CAD \cong \angle CBD$, and $AD \cong BD$, then $\angle 1 = \angle ACD \cong \angle BCD = \angle 2$. □
- 5) *Proof.* In figure 6-36, if $\angle ABC \cong \angle ABE$ and $CB \cong BE$, then $\angle C \cong \angle E$.
1. Assume $\angle ABC \cong \angle ABE$ and $CB \cong BE$.
 2. Observe that since $\angle ABC \cong \angle ABE$, then their supplementary angles $\angle CBD$ and $\angle EBD$ are congruent.
 3. Since $CB \cong EB$, $\angle CBD \cong \angle EBD$, and $BD \cong BD$, then $\angle C \cong \angle E$ by S.A.S. postulate. □
- 6) *Proof.* In figure 6-37, if $AC \cong DF$, $BD \cong CE$, and $\angle 1 \cong \angle 2$, then $\angle A \cong \angle F$.
1. Assume $AC \cong DF$, $BD \cong CE$, and $\angle 1 \cong \angle 2$.
 2. Since $AC \cong FD$ and $CD \cong DC$, then $AD \cong FC$.
 3. Since $AD \cong FC$, $\angle 2 \cong \angle 1$, and $DB \cong CE$, then $\angle A \cong \angle F$ by S.A.S. postulate. □
- 7) *Proof.* In figure 6-38, if $AD \cong EC$ and $BE \cong BD$, then $\angle A \cong \angle C$.
1. Assume $AD \cong EC$ and $BE \cong BD$.
 2. Since $BE \cong BD$ then $\triangle EBD$ is an isosceles triangle with base angles $\angle BED \cong \angle BDE$.
 3. Since $\angle BED \cong \angle BDE$, then their supplementary angles $\angle BEA$ and $\angle BDC$ are congruent.
 4. Since $AD \cong EC$ and $ED \cong DE$, then $AE \cong DC$.
 5. Since $BE \cong BD$, $\angle BEA \cong \angle BDC$, and $AE \cong CD$, then $\angle A \cong \angle C$ by S.A.S. postulate. □
- 8) *Proof.* In figure 6-39, if $RS \cong RT$ and $US \cong UT$, then $\angle RSU \cong \angle RTU$.
1. Assume $RS \cong RT$ and $US \cong UT$.
 2. Since $RS \cong RT$ then $\triangle SRT$ is an isosceles triangle with base angles $\angle RST \cong \angle RTS$.
 3. Since $US \cong UT$ then $\triangle SUT$ is an isosceles triangle with base angles $\angle UST \cong \angle UTS$.
 4. Since $\angle RST \cong \angle RTS$ and $\angle UST \cong \angle UTS$, then $\angle RSU \cong \angle RTU$. □
- 9) *Proof.* In figure 6-40, if $AC \cong BC$ and $CD \cong CE$, then $\angle 1 \cong \angle 2$.
1. Assume $AC \cong BC$ and $CD \cong CE$.
 2. Observe that $\angle C \cong \angle C$.
 3. Since $AC \cong BC$, $\angle C \cong \angle C$, and $CE \cong CD$, then $\angle 1 \cong \angle 2$ by S.A.S. postulate. □



- 10) *Proof.* In figure 6-41, if $AO \cong OD$, $AF \cong FD$, $BO \cong OC$, and $BF \cong FC$, then $\angle EAB \cong \angle EDC$.
1. Assume $AO \cong OD$, $AF \cong FD$, $BO \cong OC$, and $BF \cong FC$.
 2. Since $OB \cong OC$ then $\triangle BOC$ is an isosceles triangle with base angles $\angle OBF \cong \angle OCF$.
 3. Since $OB \cong OC$, $\angle OBF \cong \angle OCF$, and $BF \cong CF$, then $\angle OFB \cong \angle OFC$ by S.A.S. postulate.
 4. Since $EF \cong EF$, $\angle EFA \cong \angle EFD$, and $FA \cong FD$, then $\angle EAB \cong \angle EDC$ by S.A.S. postulate. □



- 11) *Proof.* In figure 6-41, if $AC \cong BC$, $CG \cong CF$, and $\angle 1 \cong \angle 2$, then $\angle 3 = \angle CGE \cong \angle CFA = \angle 4$.
1. Assume $AC \cong BC$, $CG \cong CF$, and $\angle 1 \cong \angle 2$.
 2. Since $\angle 1 = \angle ACD \cong \angle BCE = \angle 2$ and $\angle GCF \cong \angle FCG$, then $\angle ACF \cong \angle BCG$.
 3. Since $AC \cong BC$, $\angle ACF \cong \angle BCG$, and $CF \cong CG$, then $\angle CGE \cong \angle CFA$ by S.A.S. postulate. □



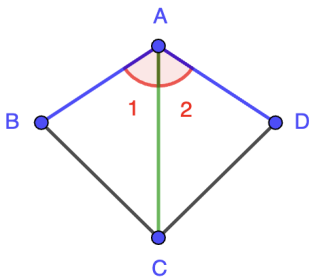
Exercise Group 6.7

Proof. If $\triangle ABC \cong \triangle A'B'C'$ and $\triangle A'B'C' \cong \triangle A''B''C''$, then $\triangle ABC \cong \triangle A''B''C''$.

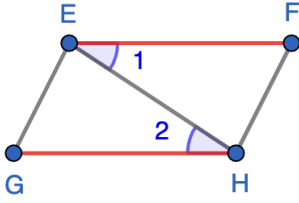
1. Assume $\triangle ABC \cong \triangle A'B'C'$ and $\triangle A'B'C' \cong \triangle A''B''C''$.
2. Since $\triangle ABC \cong \triangle A'B'C'$, then $AB \cong A'B'$, $AC \cong A'C'$, $BC \cong B'C'$, $\angle A \cong \angle A'$, $\angle B \cong \angle B'$, and $\angle C \cong \angle C'$.
3. Since $\triangle A'B'C' \cong \triangle A''B''C''$, then $A'B' \cong A''B''$, $A'C' \cong A''C''$, $B'C' \cong B''C''$, $\angle A' \cong \angle A''$, $\angle B' \cong \angle B''$, and $\angle C' \cong \angle C''$.
4. Then since segment and angle congruence are transitive, we get $AB \cong A''B''$, $AC \cong A''C''$, $BC \cong B''C''$, $\angle A \cong \angle A''$, $\angle B \cong \angle B''$, and $\angle C \cong \angle C''$.
5. Therefore $\triangle ABC \cong \triangle A''B''C''$. □

Exercise Group 6.8

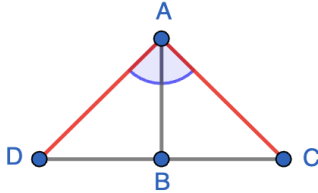
- 1) *Proof.* In figure 6-44, if $AB \cong BC$ and $\angle 1 \cong \angle 2$, then $AD \cong DC$.
1. Assume $AB \cong BC$ and $\angle 1 \cong \angle 2$.
 2. Observe that $BD \cong BD$.
 3. Since $AB \cong CB$, $\angle 1 \cong \angle 2$, and $BD \cong BD$, then $\triangle ABD \cong \triangle CBD$ by S.A.S.
 4. Therefore $AD \cong CD$. □



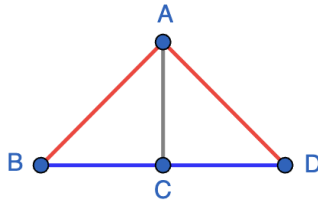
- 2) *Proof.* In figure 6-45, if $EF \cong GH$ and $\angle 1 \cong \angle 2$, then $\triangle GHE \cong \triangle FEH$.
1. Assume $EF \cong GH$ and $\angle 1 \cong \angle 2$.
 2. Observe that $EH \cong HE$.
 3. Since $FE \cong GH$, $\angle 1 \cong \angle 2$, and $EH \cong HE$, then $\triangle FEH \cong \triangle GHE$ by S.A.S. □



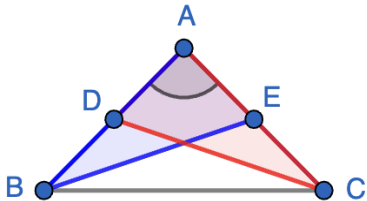
- 3) *Proof.* In figure 6-46, if $AC \cong AD$ and AB bisects $\angle A$, then $\angle CBA \cong \angle DBA$.
1. Assume $AC \cong AD$ and AB bisects $\angle A$.
 2. Since AB bisects $\angle A$, then $\angle CAB \cong \angle DAB$.
 3. Since $CA \cong DA$, $\angle CAB \cong \angle DAB$, and $AB \cong AB$, then $\triangle CAB \cong \triangle DAB$ by S.A.S.
 4. Therefore $\angle CBA \cong \angle DBA$. □



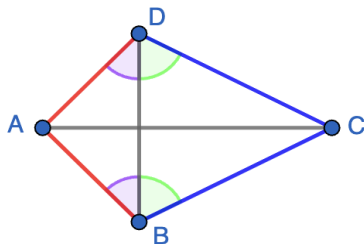
- 4) *Proof.* In figure 6-47, if $AB \cong AD$ and $BC \cong CD$, then $\triangle ABC \cong \triangle ADC$.
1. Assume $AB \cong AD$ and $BC \cong CD$.
 2. Observe that $\triangle ABD$ is isosceles, and so base angles $\angle ABC$ and $\angle ADC$ are congruent.
 3. Since $AB \cong AD$, $\angle ABC \cong \angle ADC$, and $BC \cong DC$, then $\triangle ABC \cong \triangle ADC$ by S.A.S. □



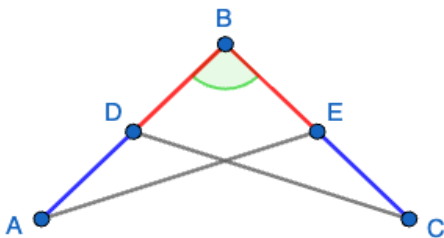
- 5) *Proof.* In figure 6-48, if $AB \cong AC$, D is the midpoint of AB , and E is the midpoint of AC , then $\triangle ABE \cong \triangle ACD$.
1. Assume $AB \cong AC$, D is the midpoint of AB , and E is the midpoint of AC .
 2. Since D is the midpoint of AB and E is the midpoint of AC , then $\overline{AD} = \frac{1}{2}\overline{AB} = \frac{1}{2}\overline{AC} = \overline{AE}$, and so $AD \cong AE$.
 3. Since $DA \cong EA$, $\angle A \cong \angle A$, and $AC \cong AB$, then $\triangle DAC \cong \triangle EAB$ by S.A.S. □



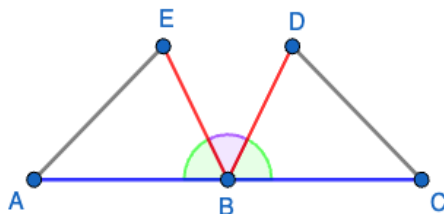
- 6) *Proof.* In figure 6-49, if $AB \cong AD$ and $BC \cong CD$, then $\angle ABC \cong \angle ADC$ and $\angle BAC \cong \angle DAC$.
1. Assume $AB \cong AD$ and $BC \cong CD$.
 2. Observe that $\triangle CDB$ and $\triangle ADB$ are both isosceles, and so $\angle CDB \cong \angle CBD$ and $\angle ADB \cong \angle ABD$.
 3. Then by postulate V-3, we know that $\angle ADC \cong \angle ABC$, and it remains to show that $\angle BAC \cong \angle DAC$.
 4. Since $AB \cong AD$, $\angle ABC \cong \angle ADC$, and $BC \cong DC$, then $\triangle ABC \cong \triangle ADC$ by S.A.S.
 5. Therefore $\angle BAC \cong \angle DAC$. □



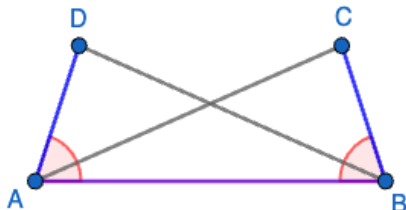
- 7) *Proof.* In figure 6-50, $AB \cong BC$ and D, E are the midpoints of AB and BC respectively. Prove that $AE \cong CD$.
1. Assume $AB \cong BC$ and D, E are the midpoints of AB and BC respectively.
 2. Observe that $AB \cong CB$, $\angle B \cong \angle B$, and $BE \cong BD$.
 3. Then $\triangle ABE \cong \triangle CBD$ by S.A.S.
 4. Therefore $AE \cong CD$. □



- 8) *Proof.* In figure 6-51, $EB \cong BD$, B is the midpoint of AC , and $\angle EBC \cong \angle DBA$. Prove that $AE \cong CD$.
1. Assume $EB \cong BD$, B is the midpoint of AC , and $\angle EBC \cong \angle DBA$.
 2. Observe that $AB \cong CB$, $\angle ABE \cong \angle CBD$ (equals subtracted from equals are equal), and $BE \cong BD$.
 3. Then $\triangle ABE \cong \triangle CBD$ by S.A.S.
 4. Therefore $AE \cong CD$. □

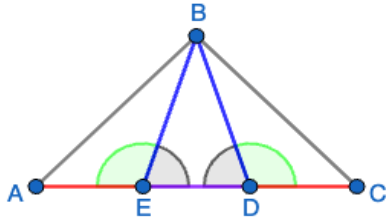


- 9) *Proof.* In figure 6-52, $\angle DAB \cong \angle CBA$ and $AD \cong BC$. Prove that $BD \cong AC$.
1. Assume $\angle DAB \cong \angle CBA$ and $AD \cong BC$.
 2. Observe that $DA \cong CB$, $\angle DAB \cong \angle CBA$, and $AB \cong BA$.
 3. Then $\triangle DAB \cong \triangle CBA$ by S.A.S.
 4. Therefore $BD \cong AC$. □



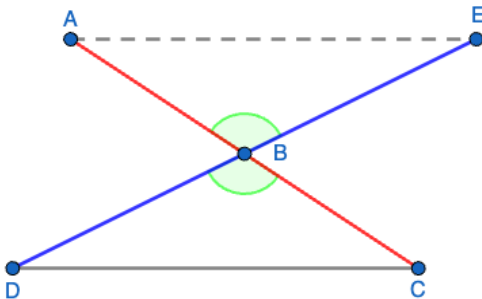
- 10) *Proof.* In figure 6-53, $AD \cong EC$, and $BE \cong BD$. Prove that $\triangle ABC$ is isosceles.
1. Assume $AD \cong EC$ and $BE \cong BD$.
 2. Observe that $\angle BED \cong \angle BDE$ since $\triangle BED$ is isosceles.
 3. Then $\angle AEB \cong \angle CDB$ since their supplements are congruent, and $EA \cong DC$.
 4. Since $BE \cong BD$, $\angle BEA \cong \angle BDC$, and $EA \cong DC$, then $\triangle BEA \cong \triangle BDC$ by S.A.S.
 5. Therefore $BA \cong BC$, and so $\triangle ABC$ is isosceles.

□



- 11) *Proof.* Assume $\overline{AB} = \overline{CB}$, $\overline{DB} = \overline{EB}$, and $\overline{CD} = 78$ ft. Then $\overline{AE} = 78$ ft.
1. Observe that $\angle ABE \cong \angle CBD$ since they're vertical angles.
 2. Observe that $AB \cong CB$ and $DB \cong EB$.
 3. Since $AB \cong CB$, $\angle ABE \cong \angle CBD$, and $DB \cong EB$, then $\triangle ABE \cong \triangle CBD$ by S.A.S.
 4. Then $AE \cong CD$, and so they are the same length.
 5. Therefore $\overline{AE} = 78$ ft.

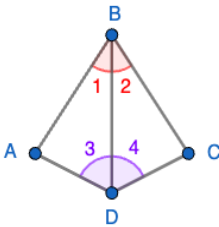
□



Exercise Group 6.9

- 1) *Proof.* In figure 6-56, if $\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$, then $\triangle ABD \cong \triangle CBD$.
1. Assume $\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$.
 2. Observe that $BD \cong BD$.
 3. Since $\angle 1 \cong \angle 2$, $BD \cong BD$, and $\angle 3 \cong \angle 4$, then $\triangle ABD \cong \triangle CBD$ by A.S.A.

□



- 2) *Proof.* In figure 6-57, if $\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$, then $AB \cong CD$.
1. Assume $\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$.
 2. Observe that $BD \cong DB$.
 3. Since $\angle 1 \cong \angle 2$, $BD \cong DB$, and $\angle 3 \cong \angle 4$, then $\triangle ABD \cong \triangle CDB$ by A.S.A.
 4. Therefore $AB \cong CD$

□

